

AI-Enhanced Image Processing and Management Web Application (SRS)

Group 2 Senior Project
Supervisor: Dr. Faisal Arafsha

Name	ID
Abdulrahman Alguydi	YB0062
Mohammed Dafterdar	VA0056
Saud Almansour	VA0040

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1. Introduction

1.1 Document Purpose

The purpose of this Software Requirements Specification (SRS) is to provide a comprehensive technical blueprint for the development of a full-stack image processing and storage platform. This document serves as the definitive reference for all functional and non-functional requirements, ensuring that the final system aligns with the intended design goals. It is intended for project evaluators and developers to verify that all image manipulation tools, AI-driven enhancements, and storage protocols are implemented with technical accuracy and system integrity.

1.2 Project Scope

The objective of this project is to develop a centralized web-based ecosystem that empowers users to manage, enhance, and transform digital imagery. The system provides value by offering a dual-track editing suite: a collection of standard non-AI presets for geometric and color manipulation, alongside sophisticated AI-powered tools for super-resolution upscaling and automated color correction.

The project focus is on delivering high-quality image outputs and a seamless user experience through an interactive "Before vs. After" comparison slider. While the system allows users to download their modified assets, security features are considered optional and will not be implemented in this version of the project. The development is scheduled for completion within the second half of 2026.

1.3 Abstract

A full-stack web application designed for the secure storage and intelligent modification of digital images. The platform provides a centralized environment where users can store and view their personal photo libraries while utilizing a suite of built-in editing tools. These tools include standard geometric transformations such as cropping, flipping, and swirl effects as well as compression and color adjustments. Additionally, the system integrates AI-powered enhancements, including upscaling and automated color correction to restore low-quality images. To ensure a seamless user experience, an interactive comparison slider allows for real-time, side-by-side visualization of all modifications against the original source files.

2. General Description

2.1 Product Perspective

The project is a standalone, full-stack web application that integrates traditional image processing logic with advanced AI-driven enhancement models. It operates as an independent ecosystem consisting of a client-side user interface, a backend server for processing, and a centralized database for asset management. It utilizes external Python-based libraries for image transformation and AI upscaling but if that was found to be difficult to implement locally, an API for online editing will be used instead.

2.2 Product Features

The system provides several core features designed to give users a comprehensive image management experience:

- **Centralized Image Storage:** Users can upload and securely organize their personal photo libraries in one online location.
- **Standard Editing Presets:** A suite of non-AI tools allows for quick geometric transformations such as crop, flip, and swirl alongside color adjustments.
- **AI-Powered Enhancements:** Advanced tools like upscaling and automated color correction transform low-quality images into high-quality assets.
- **Interactive Comparison Slider:** A real-time visual tool that allows users to compare their modifications against the original image side-by-side.
- **Exporting and Downloading:** Users can download the final, modified high-quality images to their local devices for external use.

2.3 User Classes and Characteristics

The user community for this platform is divided into two primary classes based on their editing needs and technical objectives:

- **Simple Users (Guests):** This class consists of individuals seeking a quick and accessible way to perform standard image modifications. Their primary characteristics include a preference for non-AI editing tools—such as cropping, flipping, and swirl effects—to make rapid adjustments to their stored photos without requiring advanced technical knowledge.
- **Advanced Users (Registered):** This class represents individuals who require high-quality image restoration and enhancement. These users specifically utilize the AI-powered upscaling and color correction features alongside the centralized storage system to manage and improve low-quality digital assets for professional or personal use.

2.4 Operating Environment

The application operates within a distributed web-based infrastructure consisting of three main components:

- **Client-Side:** Users access the platform via modern web browsers (such as Chrome, Firefox, or Safari). The environment must support JavaScript and CSS to enable the interactive comparison slider and account dashboard.

- **Server-Side:** The backend environment must be capable of executing Python-based image processing libraries and hosting the AI models required for image upscaling and color correction.
- **Database Layer:** The system requires a centralized database to store user account credentials, image metadata, and the relational links between original and modified files. This ensures that Registered Users can persistently access their personal photo libraries.

2.5 Design and Implementation Constraints

The following restrictions govern the design and implementation of the platform:

- **Security Constraint:** Building advanced security protocols or data encryption is not a requirement for this project; these are considered optional features and will not be implemented in the current version.
- **Resolution Constraint:** The system is restricted to processing images with a maximum resolution of 1080p to maintain stable performance and prevent system overloads during AI transformations.
- **Output Constraint:** The system must ensure that the quality of the processed image is at least equal to the quality of the original input file, preventing any loss of fidelity during geometric transformations or AI upscaling.
- **Software Constraint:** The application is restricted to supporting standard digital image formats, specifically JPG and PNG, to ensure reliability across all editing functions.
- **Resource Constraint:** The application must operate within specific server-side memory and processing limits to manage the computational load of AI-driven presets effectively.

2.6 Assumptions and Dependencies

The successful development and operation of the platform rely on the following assumptions and dependencies:

- **Internet Connectivity:** It is assumed that users have a stable internet connection to facilitate the uploading of original files and the downloading of high-quality modified assets.
- **Input Integrity:** The system assumes that users will provide valid, uncorrupted image files in the supported JPG or PNG formats.
- **Library Dependencies:** The intelligent modification engine depends on the continued availability and compatibility of external Python-based libraries for image manipulation and AI-driven upscaling.
- **System Availability:** The platform depends on a functional and active database service to manage user accounts and maintain the relational links between original and modified images.
- **Hardware Capability:** It is assumed that the server-side environment provides sufficient computational resources to execute AI presets without exceeding the performance constraints of the web application.

3. System Features

3.1 Image Upload and Initial Processing

Description

This feature allows users to bring digital images into the application workspace from their local hardware. To streamline development, the system uses a temporary session to hold the image initially. Permanent storage and account linking only occur if the user chooses to modify the image.

Input and Output

- **Input:** The user selects a digital image file and clicks the "Upload" button.
- **Valid Input Range:** Files must be in JPG or PNG format with a maximum resolution of 1080p.
- **Output:** The image is loaded into the web editor's temporary session memory for immediate viewing.

Functional Requirements

- **REQ 3.1-1:** The system shall validate that the input file is a supported format (JPG or PNG).
- **REQ 3.1-2:** The system will block any upload that exceeds the 1080p resolution limit to protect server performance.
- **REQ 3.1-3:** For Registered Users, the system shall only save the image to the permanent database and link it to the user's profile once an edit tool is applied.
- **REQ 3.1-4:** For Guest Users, the system shall maintain the image only in temporary session memory. The system shall not interact with the database for Guest User uploads.
- **REQ 3.1-5:** If the upload fails due to format or resolution errors, the system shall provide an error message and clear the temporary session.

3.2 Standard Editing Presets

Description

This section defines the system's expected behavior and the specific outputs produced based on user inputs. These features describe the direct relationship between what a user enters and the resulting modified image.

Input and Output

- **Input:** The user selects a manual tool and provides necessary parameters, such as a crop region or a flip orientation.
- **Valid Input Range:** Images must be in JPG or PNG format with a maximum resolution of 1080p.
- **Output:** The system performs the requested data processing to generate a modified version of the image.

Functional Requirements

- **REQ 3.2-1:** The system shall apply geometric transformations, including cropping, flipping, and swirl effects, based on user-provided inputs.
- **REQ 3.2-2:** The system shall adjust image colors and brightness according to the parameters specified by the user.

- **REQ 3.2-3:** The system must ensure that the output image maintains a quality level at least equal to the original source file.
- **REQ 3.2-4:** For Registered Users, applying a preset shall trigger the system to save both the original and modified images to the database.
- **REQ 3.2-5:** For Guest Users, the modified output must be displayed in the session workspace without a database record being created.

3.3 AI-Powered Enhancements

Description

This section covers advanced modifications that use artificial intelligence to restore or improve image quality. These requirements specify how inputs are used to produce high-quality outputs.

Input and Output

- **Input:** The user selects an AI-driven enhancement tool, such as upscaling or color correction.
- **Valid Input Range:** Input files must be in JPG or PNG format and cannot exceed a 1080p resolution.
- **Output:** The system produces an enhanced version of the image that improves upon the original source.

Functional Requirements

- **REQ 3.3-1:** The system shall use AI-powered upscaling to increase the clarity and detail of the input images if this option is chosen by the user.
- **REQ 3.3-2:** The system shall apply automated color correction to optimize the balance and tone of the digital file if this option is chosen by the user.
- **REQ 3.3-3:** For Registered Users, applying any AI enhancement shall trigger the system to save both the original and the new high-quality output to the database.
- **REQ 3.3-4:** The system must ensure the output quality is at least equal to or better than the original input file.

3.4 Interactive Comparison Slider

Description

This feature provides a visual tool that allows users to compare the original image with the modified version side-by-side. It helps the user conduct a visual audit of the changes to ensure the results meet their expectations.

Input and Output

- **Input:** The user interacts with a slider handle on the screen, moving it horizontally across the image workspace.
- **Valid Input Range:** The slider is active only after at least one modification has been applied to a valid JPG or PNG file.
- **Output:** The system dynamically updates the display to show the original file on one side of the slider and the modified version on the other.

Functional Requirements

- **REQ 3.4-1:** The system shall display an interactive slider over the image workspace if this view is chosen by the user.
- **REQ 3.4-2:** The system must show the original "before" image and the modified "after" image simultaneously, separated by the slider handle.
- **REQ 3.4-3:** The system shall update the visual split between the two images in real-time as the user moves the slider handle.
- **REQ 3.4-4:** The system must ensure that the slider remains functional for all supported modifications, including cropping, swirling, and AI upscaling.
- **REQ 3.4-5:** In cases where the aspect ratio changes (e.g., after cropping), the system must center both images within a consistent workspace to maintain a smooth visual comparison.

3.5 Image Gallery Management

Description

This feature allows Registered Users to view and manage their personal collection of saved images. It provides a persistent storage environment, allowing users to access their original and modified photos at any time.

Input and Output

- **Input:** The user navigates to their personal gallery or selects a specific image to delete.
- **Valid Input Range:** This feature is only accessible to Registered Users with a valid account.
- **Output:** The system displays a collection of the user's saved images or removes a selected file from the database.

Functional Requirements

- **REQ 3.5-1:** The system shall retrieve and display all images linked to the user's specific account from the database.
- **REQ 3.5-2:** The system must allow the user to delete a specific image from their gallery if this action is chosen by the user.
- **REQ 3.5-3:** Upon a delete command, the system must permanently remove the image file and its associated data from the database.
- **REQ 3.5-4:** The system must ensure that a user can only view or manage images that are linked to their own account.
- **REQ 3.5-5:** The system shall allow the user to select any image from their gallery to reload it into the editing workspace for further modification, using the original image as the input.

4. External Interface Requirements

4.1 User Interfaces

The interface is designed as a streamlined, single-page web application that focuses on ease of use. The layout follows these standards:

- **Global Navigation Bar:** A persistent bar at the top of the page that always remains visible.
 - **Profile Icon and Username:** Located in the top-right corner, this icon allows Registered Users to log in, view their account details, or access their personal gallery of previously uploaded and edited images.
 - **Gallery:** Located to the left of the profile icon for quick access to saved projects.
 - **Upload Button:** Located next to the gallery button with prominent colors; it allows the user to upload pictures even if they have left the initial upload/home page.
 - **Logo:** Website Logo sits in the left of the bar.
- **Central Upload Hub:** The center of the landing page features a prominent box that supports both clicking to upload and a drag-and-drop function for image files. It will show the uploaded image's metadata underneath the it.
- **Feature Toggle:** Positioned directly beneath the upload area is a wide toggle switch. This allows the user to flip between Standard Options and AI Options.
- **Dynamic Tool Bar:** Based on the toggle selection, a horizontal row of buttons appears with intuitive emojis and labels:
 - **Standard Mode:**  Crop,  Flip,  Swirl, and  Color Adjust.
 - **AI Mode:**  AI Upscale and  AI Color Fix.
 - **Interactive Tooltips:** The system shall display a brief text description (e.g., "Swirl Effect") when a user hovers their cursor over any icon to ensure the interface is intuitive.
- **Interactive Workspace:** Once an image is active, the central area displays the photo with the Interactive Comparison Slider, allowing for real-time visual audits of changes.
 - **Processing Feedback:** The system shall display a visual loading indicator over the workspace while any modification is being calculated. This informs the user that the system is busy and prevents multiple clicks.
- **Action Buttons:** A prominent download button is in the bottom section of the workspace, allowing users to save their results immediately to their local device.
- **Standardized Alerts:** Red or green notification banners appear at the top of the screen to inform users of errors (like exceeding the 1080p limit) or successful saves.

4.2 Hardware Interfaces

Client-Side Hardware

The application is a web-based platform designed specifically for Personal Computers. To ensure a smooth experience with the Interactive Comparison Slider and image rendering, the following hardware is required:

- **System Memory:** A minimum of 4GB RAM is required for stable browser performance during image editing.
- **Display Resolution:** A minimum resolution of 1920x1080 is required to view images and the workspace correctly without visual distortion.

- **Input Devices:** Support for standard mouse and keyboard or touch pad is required for interacting with the drag-and-drop box and the slider handle.

Server-Side Hardware

Because the application performs intensive AI-powered upscaling and color correction, the backend server must meet high hardware standards:

- **Memory (RAM):** A minimum of 32GB RAM is required to handle the loading of AI models and the concurrent processing of multiple image transformations.
- **Processor (CPU):** High-performance multi-core processors are necessary to manage data transformations and multiple user sessions.
- **Storage:** High-speed Solid-State Drives (SSD) are required to ensure fast read/write speeds when retrieving original images from the database for **REQ 3.5-5**.

Network Interface

The software requires an active internet connection to communicate between the user's computer and the backend server:

- **Connection Type:** Support for standard Ethernet, Wi-Fi, and mobile data (3G, 4G, and 5G).
- **Bandwidth:** A stable connection is needed to handle the upload and download of 1080p image files without excessive waiting times.

4.3 Software Interfaces

Client-Side Software

The application is a web-based tool and must be compatible with modern desktop environments and web standards.

- **Web Browsers:** The application must be fully functional on the latest versions of Google Chrome, Mozilla Firefox, Apple Safari, and Microsoft Edge.
- **Operating Systems:** Compatibility is required for desktop environments running Windows 10/11 and macOS.

Database Management System

To manage user accounts and image storage as defined in Section 3, the application requires a persistent relational database:

- **Database Engine:** MySQL will be the exclusive database engine used to store user credentials, image metadata, and the relational links between original and modified files.
- **Storage Interface:** The system must interface with the MySQL server to perform all "Save," "Retrieve," and "Delete" actions.

Image Processing and AI Libraries

The core functionality relies on specific backend libraries for image manipulation and enhancement:

- **Python Environment:** The backend will utilize Python as the primary programming language for processing logic.
- **Standard Image Manipulation:** Libraries such as OpenCV or Pillow will be used for geometric transformations (Crop, Flip, Swirl).
- **AI Frameworks (Local):** AI-powered enhancements (Upscaling and Color Correction) will be implemented using TensorFlow or PyTorch.
- **Fallback AI Interface (API):** If local AI implementation faces development difficulties, the system shall interface with the OpenAI API (DALL-E or similar image models) to perform the required image enhancements.

Web Frameworks

- **Backend Interface:** A Python-based web framework (such as Flask or FastAPI) will manage the communication between the processing engine and the user's browser.
- **Frontend Technologies:** Standard web technologies (HTML5, CSS3, JavaScript) will be used to build the interface, the Feature Toggle, and the Interactive Comparison Slider.

5. Nonfunctional Requirements

5.1 Performance Requirements

Response Time

The system must meet the following time constraints to provide a functional editing experience:

- **Page Load:** The initial landing page, including the Central Upload Hub, shall load within 5 seconds on a standard internet connection.
- **Standard Edits:** Manual transformations—including Crop, Flip, and Swirl—must be processed and reflected in the workspace within 5 seconds of the user's action.
- **AI Processing:** Due to the complexity of the models, AI Upscaling and Color Fix must complete processing and display the output within 15 seconds.
- **Slider Responsiveness:** The Interactive Comparison Slider must update the visual split in real-time with less than 100ms of lag during user interaction.

Throughput and Capacity

- **Concurrent Sessions:** The server infrastructure (32GB RAM) shall support 5 to 10 concurrent users performing active edits without exceeding the response times listed above.
- **Data Handling:** The system must be optimized to handle the upload, processing, and download of 1080p images (JPG/PNG) without system timeouts.

Resource Utilization

- **Session Management:** Temporary session memory used for Guest Users must be purged immediately after the browser session is closed to maintain server availability.
- **Database Retrieval:** For Registered Users, the retrieval of an "Original" image from the MySQL database for a new editing session must occur within 5 seconds.

5.2 Safety and Security Requirements

Safety and security requirements define the protections the system must provide to ensure data privacy, prevent unauthorized access to user accounts, and maintain the confidentiality of images.

User Authentication

- **REQ 5.2-1:** The system shall require a unique username and a secure password for all Registered Users to access their personal gallery and account settings.
- **REQ 5.2-2:** Password data must be stored in the MySQL database using a one-way cryptographic hash to ensure that plain-text passwords are never accessible.

Data Access and Privacy

- **REQ 5.2-3:** The system must ensure that images saved in the database can only be accessed or deleted by the original uploader.
- **REQ 5.2-4:** For Guest Users, the uploaded image will not be saved server-side and will only exist for the duration of the active session.
- **REQ 5.2-5:** The system shall not share or display user-uploaded images in any public-facing gallery or feed.

System and Connection Security

- **REQ 5.2-6:** The application shall utilize HTTPS for all data transmissions to protect image files and user credentials during upload and download.
- **REQ 5.2-7:** The backend shall use parameterized queries for all MySQL interactions to protect the database from SQL injection attacks.
- **REQ 5.2-8:** If the system utilizes the OpenAI API for processing, all communication must be handled through secure, authenticated API keys stored as environment variables on the server.

5.3 Software Quality Attributes

Availability

- The system shall aim for an uptime of 99.9% during the project evaluation period.
- Because this is a web-based prototype, the application must always be accessible, except during scheduled maintenance or database updates.

Usability

- **Intuitive Design:** The application shall be designed so that a first-time user can successfully upload and edit an image within 60 seconds.
- **Visual Consistency:** All icons, tooltips, and navigation buttons must remain consistent in style and placement across the entire site.
- **Accessibility:** The interface shall use high-contrast colors and clear labels to ensure that the primary editing tools are easily distinguishable.

Maintainability

- **Modular Code:** The backend (Python) and frontend (JavaScript) code shall be written in a modular format. This allows for the system to be updated or replaced efficiently and without breaking the entire system.
- **Documentation:** All core functions in the backend logic, particularly the MySQL queries and AI API calls, must be commented and documented to allow for easy troubleshooting and future development.

Reliability

- **Consistent Output:** The system must produce predictable results for non-AI edit. For example, applying a "Flip" command multiple times must yield the exact same geometric result every time.
- **Error Recovery:** If a processing error occurs (such as an API timeout), the system must remain functional and allow the user to retry the action without needing to refresh the entire page.